

The Wrath of Gonggong

Level & gameplay design case study | Third-person, single-player RPG prototype



ICE CREATION / TRAVERSAL

Create a route across water, then reuse the same interaction under pursuit and combat pressure.

DESIGN FOCUS

Give a familiar tool new uses as the surrounding conditions change.

MY ROLE

Solo level, gameplay and systems design; technical implementation and integration.

UNREAL ENGINE 5.7 / WINDOWS 64-BIT

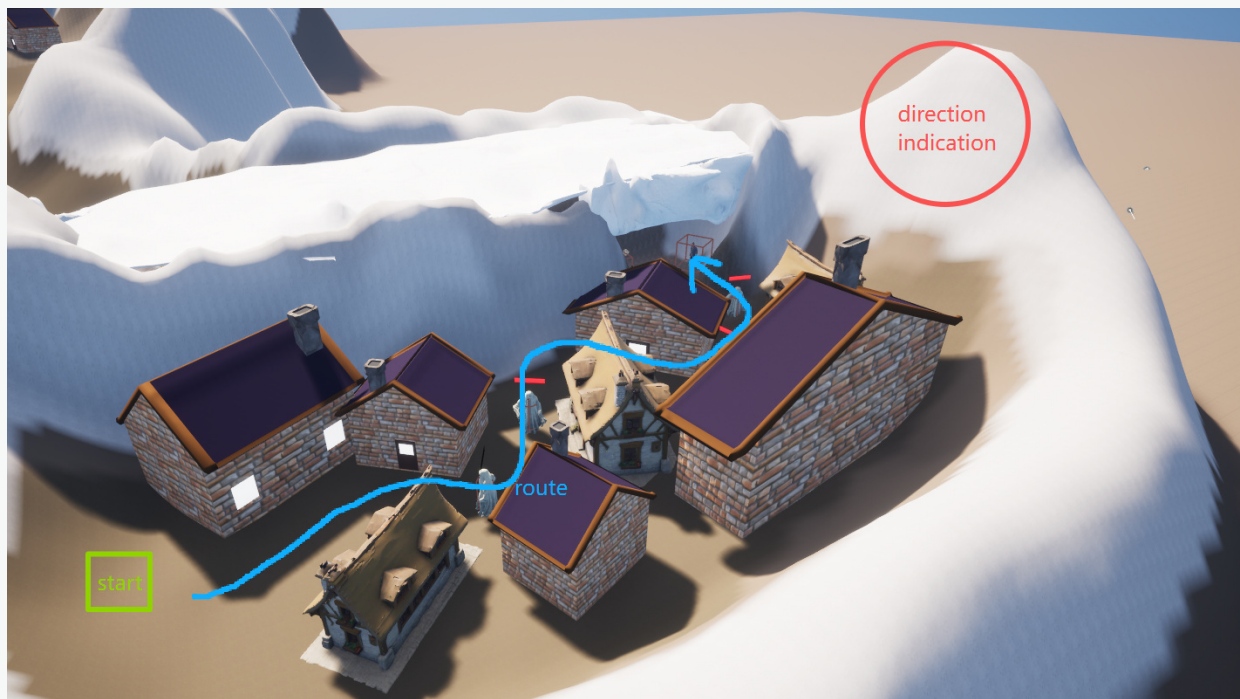
Blueprint, C++, Landscape, UMG, Sequencer and Niagara.

CURRENT EVIDENCE

3 external V0.1 testers. V0.2 developer regression completed; external retest pending.

A directed journey with local choices

Eight connected areas move from exploration and learning to escalating pressure and a final encounter.



STARTING VILLAGE / EDITOR OVERVIEW

Buildings define the passage; the distant mountain is intended as a directional landmark. Annotations show the route and key positions.

Flow reconstructed from the current build and development notes; not an original pre-production plan.

- 01 Starting village**
Movement and combat warm-up
- 02 Cave pool**
Learn ice creation; optional child rescue
- 03 Taoyuan waters**
Bridge route or an ice route across water
- 04 Taoyuan village**
Combat resolves a misunderstanding
- 05 Blizzard passage**
Reduced visibility changes navigation
- 06 Sheltered valley**
Three trolls; survivors become stronger
- 07 Avalanche and river**
Escape and cross water under pressure
- 08 Boss cave**
Combine positioning, ice and offerings

CHOICE SCOPE

Routes, rescue dialogue and collected offerings vary locally; the main storyline stays linear.

Teach once. Change the context.

The intended progression changes why, when and where the player creates ice.

01 / LEARN



Cave pool

Cross a water obstacle without combat or pursuit. Dialogue discourages wading; the exit gives a visible destination.

02 / CHOOSE



Taoyuan waters

Follow the bridges or create a direct water route. Golden offerings encourage exploration and later weaken the boss.

03 / ACT UNDER PRESSURE



Avalanche river

Reuse ice while escaping. A riverbank ice block recalls the tool; the distant cave indicates the destination.

04 / COMBINE



Boss cave

Replace lost footing as terrain collapses. Place ice while responding to attack warnings and finding openings.

Observed: all three testers used ice to cross water. This supports usability, not a measured learning curve or mastery of every stage.

Three players exposed different weaknesses

V0.1 / 28 August 2026 / Recordings and participant feedback, summarised retrospectively

PLAYER / EXPERIENCE	SESSION TIME	OBSERVATION
A / Extensive	13 min 25 sec	Enjoyed ice traversal and explored actively. Repeatedly tried to jump through the avalanche; also criticised overall polish.
B / Moderate	16 min 47 sec 10 min 38 sec (repeat)	Pulled trolls outside the arena and triggered a progression failure. Bypassed terrain on the first run; followed the intended flow on the repeat run.
C / Limited	15 min 55 sec	Used ice and finished the level, but reported unclear goals and a feeling of being pushed through the experience.

WHAT THIS EVIDENCE SUPPORTS

Two concrete iteration targets: encounter boundaries and avalanche readability. The goal-clarity feedback also identifies a future design priority.

HOW TO READ THE RESULTS

Both B runs used V0.1. The replay is not a post-change test, and the first run did not complete the intended progression. These session times are not a general completion-time benchmark.

Sample: 3 people. Qualitative findings; no claimed improvement percentage or population-level result.

Contain the observed failure path

Leaving and re-entering the encounter could prevent the final cinematic and block normal progression.



All trolls defeated; the intended transition failed.



An ice wall keeps combat inside the encounter area.

OBSERVED

Participant B pulled the trolls out, returned and killed the last troll without triggering the cinematic. The failing internal state was not isolated.

DECISION + TRADEOFF

Activate an ice wall during combat; remove it when the final cinematic begins. This blocks the observed trigger path but reduces retreat freedom.

VERIFIED + STILL OPEN

On 7 Sep 2026, developer checks in editor and packaged V0.2 covered exits, slopes, jumping and attacks. Boundaries and progression worked; external retest is pending.

Source: playertest-B1.mp4, 09:20-10:35; Troll.docx. Containment verified within tested conditions; root cause remains unresolved.

Make the visual threat match its rule

Visible gaps suggested a route through the avalanche, while a continuous hidden trigger caused failure.



Participant A attempted to jump between snowballs.

OBSERVED

A noticed the avalanche but moved towards it and repeatedly tried to pass through gaps. The recording and discussion revealed a mismatch between appearance and rule.

CHANGE + INTENT

Increase snowball size and density so the threat reads as continuous. The intended response is to escape downhill, rather than attempt to cross through it.



Larger, denser snowballs reduce visible gaps.

VERIFIED + STILL OPEN

Developer tests on 7 Sep 2026 confirmed the visual change and completable flow in editor and packaged V0.2. Whether unfamiliar players read it correctly remains untested.

Source: playertest-A.mp4, 08:58-09:16; Avalanche.docx. Next: observe unfamiliar players before explaining the intended response.

A playable prototype, with bounded conclusions

V0.2 connects the opening village to the boss and ending. Further evaluation should focus on understanding and reliability.

WHAT I WOULD TEST NEXT

01 / Threat interpretation

Observe unfamiliar players at the revised avalanche. Check their first action and explanation before offering guidance.

02 / Motivation as well as navigation

C could finish without a clear sense of purpose. Clarify the opening objective, then ask players to explain why they are moving forward.

03 / Encounter boundaries and state

B exposed a failure missed on the intended route. Retain boundary checks; investigate encounter state and reset logic if the scope grows.

CURRENT LIMIT

Strongly directed flow, limited local choices and prototype-level polish. External V0.2 retesting remains pending.

MY CONTRIBUTION

Terrain and spatial layout; gameplay systems, encounters, boss and narrative flow; UI, cinematics and Blueprint/C++ implementation. I made the design decisions and handled integration, testing and validation.

THIRD-PARTY ASSETS + AI

Character meshes are licensed assets. Environment assets are primarily licensed, arranged within my terrain and level compositions. Animation, effects, materials and textures combine licensed assets with work I created or modified.

I used AI for technical guidance and debugging. Audio and music came from AI tools or asset sources, then were selected and integrated by me.

PROJECT ORIGIN

An independent reconstruction of my design work for an unreleased commercial project. Rebuilt from the ground up without proprietary assets or confidential content; documentation rewritten.